# IMMANUVEL M

ROS 2 Robotics Developer | Mechatronics Engineering Student

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## Professional Summary

Mechatronics engineering student and ROS 2 developer who has built two complete robot-arm systems on ROS 2 Humble with MoveIt 2, ros2_control and Gazebo: webcam hand-gesture teleoperation of a 6-DOF arm, and a language-instructed pick-and-place cell with RGB-D perception and human-aware safety. Backed by hands-on hardware experience from a 12-DOF quadruped robot, embedded automation and CAD/3D printing. Seeking a ROS 2 developer or robotics software internship.

## Education

**B.E. Mechatronics Engineering** 2024 – 2028

KCG College of Technology, Chennai (Affiliated to Anna University)

## Technical Skills

| | |
|---|---|
| **ROS 2 (Humble)** | Nodes, topics, services, actions, launch files, custom messages, TF2, parameters/YAML, colcon |
| **Motion & Simulation** | MoveIt 2 (IK, OMPL, Pilz), ros2_control, Gazebo Fortress, URDF/Xacro, RViz2 |
| **Perception** | OpenCV, MediaPipe, RGB-D point clouds, camera calibration |
| **Programming** | Python, C++17, Bash, Arduino, HTML, CSS, JavaScript |
| **Testing & Tools** | pytest, gtest, Ubuntu 22.04, Git/GitHub, Claude API (LLM task planning) |
| **Embedded** | Arduino Mega, ESP32, Bluetooth, servo control, sensor integration |
| **CAD** | Fusion 360, CadQuery, Blender, Tinkercad, 3D printing, rapid prototyping |

## ROS 2 Projects

**Physical AI Workcell – Language-Instructed Pick-and-Place Robot** Individual

*ROS 2 Humble · MoveIt 2 · ros2_control · Gazebo Fortress · Python · OpenCV · Claude API*

- Built a 12-package ROS 2 system where a typed instruction (e.g. “put the red cube in the blue bin”) runs as one closed loop: perceive → plan → execute → verify, re-planning automatically when a part is moved or a grasp fails.
- Wrote class-agnostic RGB-D perception with no training data; detected 34/34 test objects with 1.5 mm mean position error.
- Implemented Pick, Place and MoveTo as ROS 2 action servers on MoveIt 2 (Pilz PTP with OMPL fallback, collision-checked straight-line approaches); task planning via the Claude API with an offline rule-based fallback.
- Added human-aware safety: speed & separation monitoring (30% speed within 0.9 m, stop within 0.45 m), latched e-stop and watchdogs; cut stop time from over 2 s to 0.28–0.36 s by commanding the controller directly.
- Benchmarked 76 randomised trials against a traditional taught cell: 84% vs 16% success, zero false success reports.

**Hand-Gesture Controlled 6-DOF Robot Arm** Individual

*ROS 2 Humble · MoveIt 2 · ros2_control · Gazebo Fortress · MediaPipe · Python · C++*

- Built an 8-package ROS 2 system in which a 6-DOF arm follows the operator’s hand from a webcam: MediaPipe tracking → gesture recognition → MoveIt 2 IK in a 20 Hz control loop → JointTrajectoryController; pinch controls the gripper.
- Modelled the arm and gripper in URDF/Xacro with joint limits and inertias; verified 125/125 workspace points reachable by IK.
- Wrote a C++ ros2_control hardware interface for a future physical arm; one launch argument switches between Gazebo, mock and real hardware (validated with a software loopback).
- Implemented a fail-safe state machine (starts disabled, thumbs-up to enable, latched e-stop, hold on lost tracking or IK failure) with motion smoothing and velocity/acceleration limits.
- Verified with 108 unit tests (pytest, gtest) and a 28-check integration test on mock and Gazebo; raised webcam rate from 8.4 to 30 FPS and, with a custom benchmark, made arm motion 63% smoother with 18% less lag.

## Hardware & Embedded Projects

**GENESIS – 12-DOF Quadruped Robot Dog** Individual

- Built a 12-DOF quadruped on Arduino Mega with MG995/MG996R servos; implemented gait algorithms for walking, turning, reverse motion and dance routines, with integrated power management for stable real-world operation.

**KIRA – Health Monitoring & Assistance Robot** Team Lead — Team of 6

- Led software for a service robot for temperature monitoring, guided assistance and user interaction; integrated sensors, embedded control and two mobile applications.

**TIFAN 2026 – Agricultural Automation System** Team Electronics Lead

- Led electronics and embedded programming for an Arduino Mega platform with conveyor, grippers and sensors, including Bluetooth communication and a mobile app for monitoring and control.

## Experience

**CAD & Additive Manufacturing Engineer (Part-Time)** EL’s Crafting — 1 Year

- Developed parametric CAD models in CadQuery, Blender and Fusion 360 for functional products.
- Ran the full 3D printing workflow from slicing to fabrication, calibration and printer maintenance.

## Achievements

- 3rd Place – Acceleration Event, GKDC Go-Kart Design Challenge, Coimbatore.
- Participant – Young Technocrats 3.0 and TIFAN 2026.